

AVISHKAR 2K26

Topic:

Analyze a real or hypothetical electrical accident caused by breaker malfunction. Suggest preventive measures based on standards and safety regulations.

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Abstract:

Circuit breakers are critical protective devices in electrical power systems, designed to automatically disconnect electrical supply during abnormal conditions such as overloads, short circuits, and earth faults. Failure or malfunction of a circuit breaker can lead to serious electrical accidents, including fires, equipment damage, and loss of human life. This abstract analyzes a real electrical accident caused by a breaker malfunction, where an electrical short circuit occurred but the circuit breaker failed to trip in time due to poor maintenance and inadequate breaking capacity. As a result, excessive fault current continued to flow, causing overheating, arcing, and fire. The investigation identified key contributing factors such as aged circuit breakers, lack of coordination between protective devices, absence of earth leakage protection, and non-compliance with electrical safety standards. Based on this analysis, preventive measures are proposed in accordance with established standards and safety regulations. These measures include proper short-circuit analysis and correct selection of circuit breakers as per IS/IEC 60947-2, regular inspection and preventive maintenance of breakers, effective coordination of protection systems, and provision of proper earthing as specified in IS 3043. Installation of RCCB or ELCB devices for earth fault protection and strict adherence to CEA Safety Regulations are also recommended. Implementation of these preventive measures can significantly reduce electrical hazards, improve system reliability, and ensure the safety of life and property.